

Q - find $P(\text{Red, sports, Imported})$ using naive Bayes

	Color	Type	origin	stolen
1	Red	sports	Domestic	yes
2	Red	sports	Domestic	no
3	Red	sports	Domestic	yes
4	yellow	sports	Domestic	no
5	yellow	sports	Imported	yes
6	yellow	SUV	Imported	no
7	yellow	SUV	Imported	yes
8	yellow	SUV	Domestic	no
9	Red	SUV	Imported	no
10	Red	sports	Imported	yes

$$P(\text{Stolen} | \text{yes}) = \frac{5}{10}$$

$$P(\text{Stolen} | \text{no}) = \frac{5}{10}$$

findings for,

$$x = P(\text{yes} | \text{red, sports, imported})$$

$$x = P(\text{no} | \text{red, sports, imported})$$

① Color:

	yes	no	$P(\text{yes})$	$P(\text{no})$
red	3	2	$3/5$	$2/5$
yellow	2	3	$2/5$	$3/5$

② Type:

	yes	no	$P(\text{yes})$	$P(\text{no})$
sports	4	2	$4/5$	$2/5$
SUV	1	3	$1/5$	$3/5$

③ Origin:

	yes	no	$P(\text{yes})$	$P(\text{no})$
Domestic	2	3	$2/5$	$3/5$
Imported	3	2	$3/5$	$2/5$

$$④ P(\text{yes} | x) = P(\text{yes} | \text{red, sports, imported})$$

$$= P(\text{yes}) * P(\text{red} | \text{yes}) * P(\text{sports} | \text{yes})$$

$$* P(\text{imported} | \text{yes})$$

$$= \frac{5}{10} * \frac{3}{5} * \frac{4}{5} * \frac{3}{5} = \boxed{0.144}$$

$$⑤ P(\text{no} | x) = P(\text{no} | \text{red, sports, imported})$$

$$= P(\text{no}) * P(\text{red} | \text{no}) * P(\text{sports} | \text{no})$$

$$* P(\text{imported} | \text{no})$$

$$= \frac{5}{10} * \frac{2}{5} * \frac{2}{5} * \frac{2}{5} = \boxed{0.032}$$

$$P(\text{yes}|\alpha) > P(\text{no}|\alpha)$$

$\therefore$ (red, sport, imported) is stolen.

find person have fever or not. 6/8 = 53/911

person	COVID (yes/no)	flu	fever.
1	yes	no	yes
2	no	yes	yes
3	yes	yes	yes
4	no	no	no
5	yes	no	yes
6	no	no	yes
7	yes	no	yes
8	yes	no	no
9	no	yes	yes
10	no	yes	no

Given : Person (flu, covid)

classify : fever yes or no as per flu & covid

$$P(\text{yes} | \text{flu}, \text{covid}) = P(\text{flu} | \text{yes}) \times P(\text{covid} | \text{yes}) \times P(\text{yes})$$

$$P(\text{no} | \text{flu}, \text{covid}) = P(\text{flu} | \text{no}) \times P(\text{covid} | \text{no}) \times P(\text{no})$$

* check for covid

	yes	no	$P(\text{yes})$	$P(\text{no})$
yes	4	1	$4/7$	$1/3$
no	3	2	$3/7$	$2/3$

$$= 7 \quad = 3 \quad = 10$$

* Check for flu

	yes	no	$P(\text{yes})$	$P(\text{no})$
yes	3	1	$3/7$	$1/3$
no	4	2	$4/7$	$2/3$

$$= 7 \quad = 3 \quad = 10$$

* Prior probability

$$P(\text{fever} = \text{yes}) = \frac{7}{10}$$

$$P(\text{fever} = \text{no}) = \frac{3}{10}$$

* conditional probability

$$\text{covid} - \frac{4}{7} \quad \frac{1}{3}$$

$$\text{flu} - \frac{3}{7} \quad \frac{2}{3}$$

⑥ finding for yes -

$$p(\text{yes} | \text{flu}, \text{covid}) = p(\text{yes}) * (p(\text{flu} | \text{yes}) * p(\text{covid} | \text{yes}))$$

$$p(\text{yes} | \text{flu}, \text{covid}) = \frac{7}{10} * \frac{1}{7} * \frac{4}{7}$$

$$= \boxed{0.17}$$

⑥ for no :-

$$p(\text{No} | \text{flu}, \text{covid}) = p(\text{no}) * p(\text{flu} | \text{no}) * p(\text{covid} | \text{no})$$

$$= \frac{3}{10} * \frac{2}{7} * \frac{2}{3} = \boxed{0.133}$$

$$p(\text{yes}) > p(\text{no})$$

∴ there fore, person have fever.